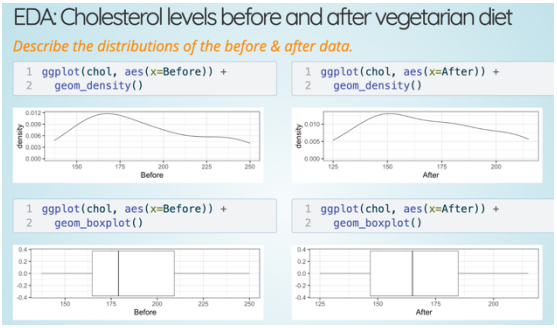
Day 10 Part 2 slides: brief answers/solutions to select questions

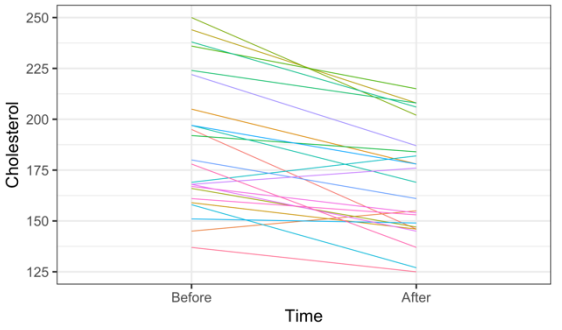
8: Come up with 2 more examples of paired study designs.



11: Describe the distributions of the before & after data.

Skewed right.
No outliers

12: What does this figure tell us?



Cholesterol levels decreased for most (but not all) people.

Individuals with higher pre-vegetarian diet cholesterol levels all appear to experience lower post-diet cholesterol levels and to a greater degree than those with lower pre-diet cholesterol levels.

13: What is this code doing?

```
chol <- chol %>%
  mutate(DiffChol = After-Before)
head(chol, 8)
```

```
# A tibble: 8 × 3
  Before After DiffChol
   <dbl> <dbl>    <dbl>
1    195   146      -49
2    145   155       10
3    205   178      -27
4    159   146      -13
5    244   208      -36
6    166   147      -19
7    250   202      -48
8    236   215      -21
```

Creates column of differences

13: Is the mean of DiffChol the same as the difference in means of After - Before? Should it be? Why or why not?

```
chol %>%
  get_summary_stats(type = "common") %>%
  gt()
```

variable	n	min	max	median	iqr	mean	sd	se	ci
Before	24	137	250	179	44.50	187.792	33.160	6.769	14.002
After	24	125	215	165	38.00	168.250	26.796	5.470	11.315
DiffChol	24	-49	13	-19	20.25	-19.542	16.806	3.430	7.096

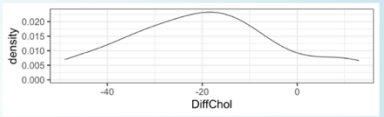
yes. yes.
Diff in means = 187.792 – 168.250 = 19.542

$\frac{\sum x_i}{n}$

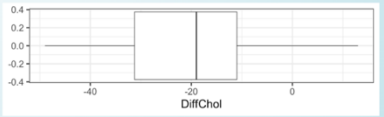
$$\frac{\sum x_i}{n} - \frac{\sum y_i}{n} = \frac{\sum x_i - y_i}{n}$$

DiffChol

```
1 ggplot(chol, aes(x=DiffChol)) +
2   geom_density()
```




```
1 ggplot(chol, aes(x=DiffChol)) +
2   geom_boxplot()
```



14: Compare and contrast the 3 distributions. Comment on shape, center, and spread.

Differences have more symmetric distribution

16:

Null and alternative hypotheses in **words**
Include as much context as possible

Null and alternative hypotheses in **symbols**

fill in the missing parts of the hypotheses.

$H_0 :\delta =$ 0
$H_A :\delta \neq$ 0

- H_0: The population mean difference in cholesterol levels after a vegetarian diet is *fill in*
- H_A: The population mean difference in cholesterol levels after a vegetarian diet is *fill in* different than zero.

17: Calculate the test statistic.

variable	n	min	max	median	iqr	mean	sd	se	ci
DiffChol	24	-49	13	-19	20.25	-19.542	16.806	3.43	7.096

$$t_{\bar{x}_d} = \frac{\bar{x}_d - \delta_0}{\frac{s_d}{\sqrt{n}}} = \frac{-19.542 - 0}{16.806/\sqrt{24}} = -5.696519$$

17: Based on the value of the test statistic, do you think we are going to reject or fail to reject H0?

Reject

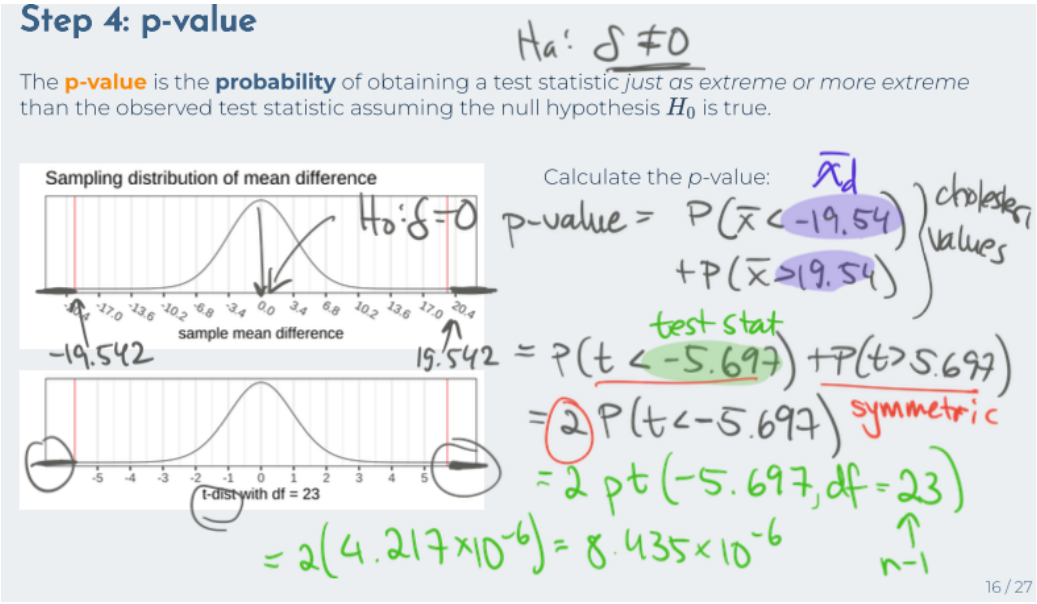
17: What probability distribution does the test statistic have?

T distribution with df = 24 – 1 = 23

17: Are the assumptions for a paired t-test satisfied so that we can use the probability distribution to calculate the p-value??

1. Differences are independent; assuming a random sample of participants
2. Nn= 24 < 30, but distribution of differences is relatively symmetric, and thus fine.

18: Calculate the p-value and shade in the area representing the p-value:

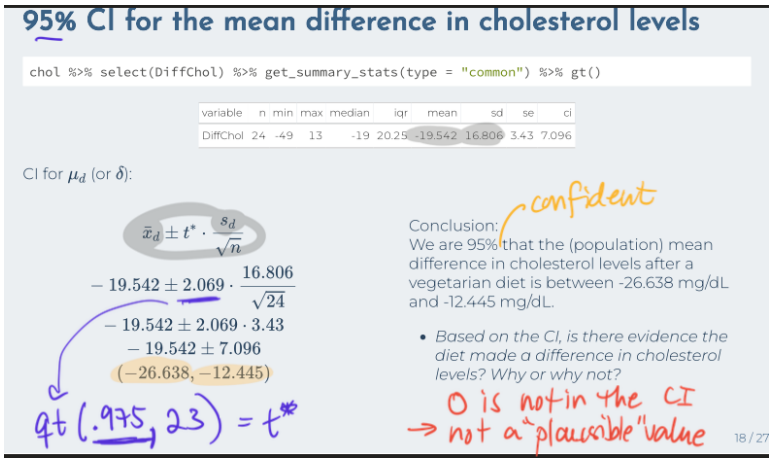


19: Do we reject or fail to reject H0?

REJECT

20: How was t∗ calculated?

qt(0.975, 23)



20: Based on the CI, is there evidence the diet made a difference in cholesterol levels? Why or why not?

Yes. Zero is not in the CI, not a plausible value.

22: Run the code without mu = 0. Do the results change? Why or why not?

NO. 0 is the default.

23: What is different in the output compared to option 1?

Option 1:

```
t.test(x = chol$DiffChol, mu = 0)
```

```
    One Sample t-test

data:  chol$DiffChol
t = -5.6965, df = 23, p-value = 8.435e-06
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -26.63811 -12.44522
sample estimates:
mean of x
-19.54167
```

Option 2:

```
t.test(x = chol$Before, y = chol$After, mu = 0, paired = TRUE)
```

```
    Paired t-test

data:  chol$Before and chol$After
t = 5.6965, df = 23, p-value = 8.435e-06
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 12.44522 26.63811
sample estimates:
mean difference
       19.54167
```

Mean and CI values are positive

25: How is the code similar and different for the 3 options?

```
# option 1
t.test(x = chol$DiffChol, mu = 0) %>% tidy() %>% gt() # tidy from broom package
```

estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
-19.54167	-5.696519	8.434775e-06	23	-26.63811	-12.44522	One Sample t-test	two.sided

```
# option 2
t.test(x = chol$Before, y = chol$After, mu = 0, paired = TRUE) %>% tidy() %>% gt(
```

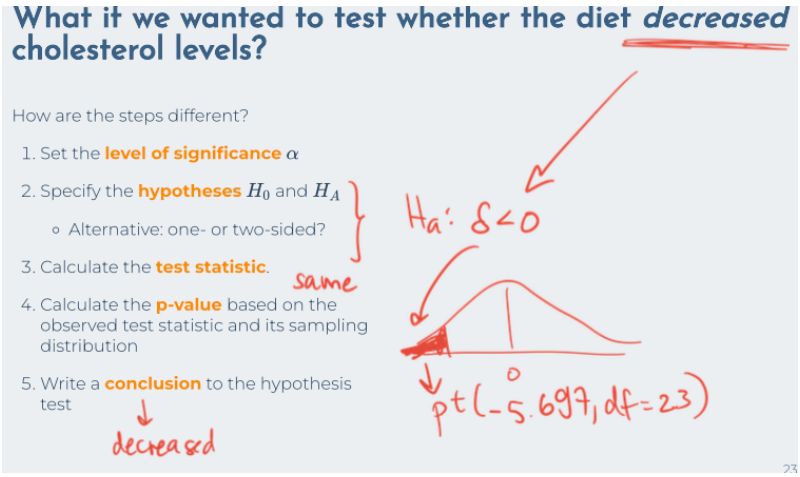
estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
19.54167	5.696519	8.434775e-06	23	12.44522	26.63811	Paired t-test	two.sided

```
# option 3
t.test(Cholesterol ~ Time, paired = TRUE, data = chol_long) %>% tidy() %>% gt()
```

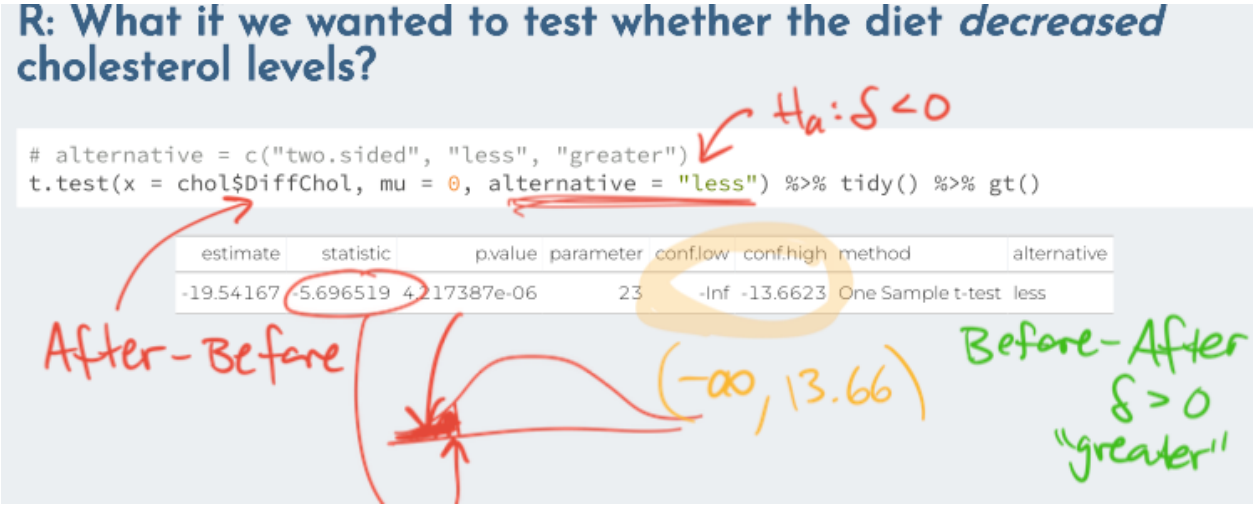
estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
19.54167	5.696519	8.434775e-06	23	12.44522	26.63811	Paired t-test	two.sided

25: Given a dataset, how would you choose which of the 3 options to use?

26: What changes in each of the steps?



27: Which of the 2 options to run a paired t-test in R is being used below?



27: How did the code change to account for testing a decrease in cholesterol levels?

See screenshot above

27: Which values in the output changed compared to testing for a change in cholesterol levels? How did they change?

See screenshot above